# WALTER VINCE T. URSUA

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Portfolio: walt1210.github.io



## CAREER PROFILE

Fourth year Computer Science student majoring in Data Science. Thesis work included building the retrieval-augmented guidance backend on Google Cloud Run and contributing to the model pipeline and Android application of an offline maize disease diagnosis system.

## SKILLS

- **Programming Languages:** Python, R, Java, C#, PHP, JavaScript, Kotlin, Dart, SQL, HTML, CSS, Assembly
- **Machine Learning and Data Science:** TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, NLTK, HDBSCAN, Matplotlib, Seaborn, Power BI
- **Agentic AI and LLM Frameworks:** LangChain (LCEL, prompt chaining, routing), CrewAI (multi-agent orchestration, tool use), Google Gemini API, Pydantic (structured LLM output), RAG pipeline design, prompt engineering, Ollama, LLM evaluation
- **Web, Mobile, and Backend Development:** Flask, Android (Kotlin), Flutter, Next.js 15, Supabase
- **Databases:** MySQL, SQLite, Oracle, ChromaDB (vector database)
- **Cloud and DevOps:** Docker, Google Cloud Run, Google Cloud Platform, Google Colab, Linux, Git, Jupyter Notebook, Visual Studio Code

## PROJECT EXPERIENCE

### MAIze: AI-Powered Maize Disease Diagnosis System (Guidance Module) *Thesis Project*

Developed a retrieval-augmented generation system that grounds AI-generated crop disease management advice in a curated agricultural knowledge base. Deployed as a cloud backend supporting a mobile diagnostic application for Filipino farmers.

**Role:** NLP and Backend Developer. Designed and deployed the retrieval-augmented generation pipeline, conducted prompt engineering, developed the evaluation methodology, and managed cloud infrastructure.

**Technologies:** Python, Flask, LangChain, Google Gemini API, ChromaDB, RAGAS, Pydantic, Docker, Google Cloud Run | **Repository:** **github.com/walt1210/maize_nlp**

### YellowMAIze: Mobile-Deployed Maize Disease Diagnosis System *Thesis Project*

Multi-task deep learning model for maize leaf disease classification, segmentation, and severity estimation, run on-device in TensorFlow Lite.

**Role:** Helped develop and improve the model pipeline, served as one of two severity raters, and worked on parts of the Android app.

**Technologies:** PyTorch, TensorFlow Lite, ONNX, scikit-learn, Grad-CAM family, Cohen's Kappa, Spearman Correlation | **Repository:** **github.com/LunocoreWired/yellowmaize**

### BayanBench: Auditing AI Knowledge of Philippine Towns *Independent Project*

Built a benchmark of 7,441 questions from the official Philippine Standard Geographic Code (1,642 cities and municipalities, 42,010 barangays) to test whether AI models answer less accurately about small and remote towns, and tested a retrieval fix using official records.

**Role:** Sole developer. Built the question set and automatic scorer (49 of 49 verdicts confirmed by hand), ran five models, and analyzed errors with logistic regression clustered by town.

**Technologies:** Python, pandas, NumPy, Ollama, Gemini API, Matplotlib | **Repository:** **github.com/walt1210/bayanbench**

### Closed-Loop Text-to-SQL Agent *Academic Project*

Developed a producer-critic reflection loop in which a Producer model drafted SQL queries and a Critic model evaluated them against a live database schema, combined with a three-pillar exception handling and recovery cascade for failed queries.

**Technologies:** Python, Google Gemini API, Pydantic, SQLite | **Repository:** **github.com/walt1210/text-to-sql-agent**

## EDUCATION

Angeles University Foundation, Bachelor of Science in Computer Science, Major in Data Science (Expected Graduation: 2027)

## CERTIFICATIONS

- Reinventing Learning through Artificial Intelligence, Holy Angel University
- Introduction to Cybersecurity, Cisco Networking Academy
- Introduction to Data Science, Cisco Networking Academy
- Microsoft Azure Fundamentals (AZ-900)
- Discovering Entrepreneurship, Cisco Networking Academy
- CCNA: Introduction to Networks, Cisco Networking Academy
- AI Prompt Design, Straits Interactive

**Interests:** Chess, language learning, photography, music, stargazing

## PORTFOLIO

### MAIze: AI-Powered Maize Disease Diagnosis System (Guidance Module) *Thesis Project*

Farmers need fast, reliable answers to crop disease questions, but general-purpose AI chatbots can generate incorrect or fabricated agricultural advice. This project developed a retrieval-augmented generation module that grounds its recommendations in a curated agricultural knowledge base, deployed as the guidance backend for a mobile diagnostic application designed for Filipino farmers.

- Retrieval pipeline querying a ChromaDB vector store of verified documents before generating a response
- Evaluated with RAGAS and a domain expert review of 30 cases, rated 4.67 out of 5 overall

**Technologies:** Python, Flask, LangChain, Google Gemini API, ChromaDB, RAGAS, Pydantic, Docker, Google Cloud Run
**Link:** **github.com/walt1210/maize_nlp**

### YellowMAIze: Mobile-Deployed Maize Disease Diagnosis System *Thesis Project*

Farmers working in the field often lack reliable internet access, making diagnostic tools that depend on cloud inference impractical on site. This project designed and validated a mobile-efficient multi-task deep learning model that performs disease classification, severity estimation, and lesion segmentation from a single leaf photograph, running entirely on-device.

- Three outputs from a single forward pass in about 60 milliseconds on a CPU, via TensorFlow Lite
- Severity estimates compared with independent ratings from two team members (Spearman correlation of 0.875)

**Technologies:** PyTorch, TensorFlow Lite, ONNX, scikit-learn, Grad-CAM family, Cohen's Kappa
**Link:** **github.com/LunocoreWired/yellowmaize**

### BayanBench: Auditing AI Knowledge of Philippine Towns *Independent Project, AI Evaluation*

Government offices and businesses are adopting AI chatbots, yet no public test measured what these models know about individual Philippine towns. This benchmark uses official government data as the answer key to measure accuracy and fairness across 400 towns and five models.

- Smallest towns were placed correctly as little as 11 percent of the time; Mindanao towns had half the odds of Luzon
- Official records in the prompt raised smallest-town accuracy to 97 percent or more and nearly ended invented names

**Technologies:** Python, pandas, NumPy, Matplotlib, Ollama, Google Gemini API, Google Colab
**Link:** **github.com/walt1210/bayanbench**

### Closed-Loop Text-to-SQL Agent *Academic Project, Agentic AI*

AI-generated SQL queries can be syntactically valid yet semantically incorrect, and can fail silently when run against a real database schema. This agent addresses that by having a Producer model draft a SQL query and a Critic model evaluate it against the live schema before execution, with a structured recovery process for queries that still fail.

- Producer-critic reflection loop for self-verification before execution
- Three-pillar exception handling and recovery cascade for failed queries

**Technologies:** Python, Google Gemini API, Pydantic, SQLite
**Link:** **github.com/walt1210/text-to-sql-agent**

### Autonomous Multi-Agent Deep Research Pipeline *Academic Project, Agentic AI*

A single model asked to both research a topic and synthesize a report often performs both tasks poorly. This pipeline splits the work between two CrewAI agents: a Research Planner equipped with a web search tool, and a Synthesis Writer equipped with a custom calculator tool. Execution tracing revealed that both agents often skipped their tools and stated invented figures with confidence.

- Sequential multi-agent orchestration with full execution tracing across handoffs
- Trace-based analysis exposing tool bypass and fabricated figures behind a polished report

**Technologies:** Python, CrewAI, Google Gemini API, DuckDuckGo Search
**Link:** **github.com/walt1210/multi-agent-research-pipeline**

### Reddit Text Classification and Topic Clustering *Academic Project, Machine Learning*

Classified ten thousand Reddit posts across ten subreddits with TF-IDF and Singular Value Decomposition features, reaching eighty percent accuracy, then discovered topics with K-Means and HDBSCAN, evaluated by Adjusted Rand Index and Silhouette Score.

**Technologies:** Python, scikit-learn, HDBSCAN, pandas, NLTK
**Link:** *Notebooks available on request*

Additional projects, including a relational database design project, are available at github.com/walt1210.